IIIT SRI CITY RESEARCH EXPO 2026 · REFERENCES

Predicting Yield: Decoding Nitrogen Control of Yeast with Hybrid Mechanistic–ML Models

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Grouped by the five rows of the Nitrogen Knowledge Network (poster section 3). Purple text = the box on the poster.

SENSE

Airoldi, E. M., Miller, D., Athanasiadou, R., Brandt, N., Abdul-Rahman, F., Neymotin, B., Hashimoto, T., Bahmani, T., & Gresham, D. (2016). Steady-state and dynamic gene expression programs in *Saccharomyces cerevisiae* in response to variation in environmental nitrogen. *Molecular Biology of the Cell*, 27(8), 1383–1396. **Limitation, not source, drives genes**

Hofman-Bang, J. (1999). Nitrogen catabolite repression in *Saccharomyces cerevisiae*. *Molecular Biotechnology*, 12, 35–71. **NCR regulators**

Zhang, W., Du, G., Zhou, J., & Chen, J. (2018). Regulation of sensing, transportation, and catabolism of nitrogen sources in *Saccharomyces cerevisiae*. *Microbiology and Molecular Biology Reviews*, 82, e00040-17. **SPS + TORC1 sensing · background figure (Fig. 6)**

Nielsen, J. (2025). The power of yeast. *Yeast*, 42(12), 303–310. **Snf1 and TORC1 crosstalk**

ALLOCATE

Yu, R., Campbell, K., Pereira, R., et al. (2020). Nitrogen limitation reveals large reserves in metabolic and translational capacities of yeast. *Nature Communications*, 11, 1881. **Nitrogen reserves**

Boer, V. M., Crutchfield, C. A., Bradley, P. H., Botstein, D., & Rabinowitz, J. D. (2009). Growth-limiting intracellular metabolites in yeast growing under diverse nutrient limitations. *Molecular Biology of the Cell*, 21(1), 198–211. **Glutamine limits growth**

Zhang, Y., Li, X., Wang, J., & Chen, H. (2024). Nutrient consumption patterns of *Saccharomyces cerevisiae* and their application in fruit wine fermentation. *Fermentation*, 10(11), 539. **Preferred-first uptake**

Li, Y., Lai, Y. H., & Lu, T. (2025). Coarse-grained modeling elucidates differential metabolism of *Saccharomyces cerevisiae* under varied nutrient limitations. *ACS Synthetic Biology*, 14(5), 1523–1532. **Low N → overflow · panel ① figure**

QUANTIFY

de Leeuw, M., & Nielsen, L. K. (2026). Kinetic sampling shows the effect of medium composition on metabolic control in *Saccharomyces cerevisiae*. *Microbial Cell Factories*, 25, 76. **Control shifts by nutrient**

Minebois, R., Ortiz-Julien, A., Galeote, V., & Sanchez, I. (2025). Combined isotopic tracer and modelling approach reveals differences in nitrogen metabolism in *S. cerevisiae*, *S. uvarum* and *S. kudriavzevii* species. *Microbial Biotechnology*, 18(1), e70087. **NCR switch fitted with ^{15}N tracing**

Chongchittapiban, P., Borg, J., Waiprib, Y., Pimsamarn, J., & Tongta, A. (2016). Development of simple kinetic models and parameter estimation for simulation of recombinant human serum albumin production by *Pichia pastoris*. **Monod kinetics · panel ① equation**

MODELS

Pereira, R., Nielsen, J., & Rocha, I. (2016). Improving the flux distributions simulated with genome-scale metabolic models of *Saccharomyces cerevisiae*. *Metabolic Engineering Communications*, 3, 153–163. **Cofactor curation**

Lu, H., Li, F., Sánchez, B. J., et al. (2019). A consensus *S. cerevisiae* metabolic model Yeast8 and its ecosystem for comprehensively probing cellular metabolism. *Nature Communications*, 10, 3586. **Yeast8 + enzyme limits · panel ②**

Kocabaş, P. (2026). Two decades of genome-scale metabolic modeling for *Saccharomyces cerevisiae*. *Gazi University Journal of Science*, 39(1), 578–594. **GEMs lack N regulation · panel ② equation**

Li, F. (2021). *Constraint-based modeling of yeast metabolism and protein secretion* (Doctoral thesis). Chalmers University of Technology. ISBN 978-91-7905-557-8. **Protein energy cost · panel ② figure**

PREDICT

Valencia-Velásquez, J., Yaker-Moreno, H. A., Martínez-Guerrero, A., et al. (2025). Advancing hybrid modeling of *Saccharomyces cerevisiae* fermentation with mixed carbon sources and urea in a mini-stirred tank reactor. *Bioprocess and Biosystems Engineering*, 48, 1919–1937. **Hybrid halves error · panel ④**

Zhao, M., Zhao, S., & Liu, F. (2023). Semi-supervised hybrid modeling of the yeast fermentation process. *Machines*, 11(1), 63. **ELM hybrid · panel ④ figure and equation**

Amor, B., Béal, C., & Corrieu, G. (2025). Machine learning-based prediction of microbial growth and acidification in yogurt fermentation at industrial temperatures. *Journal of Dairy Science*, 108(7), 10102–10115. **ML growth & pH curves · panel ③**

Desai, K. M., Vaidya, B. K., Singhal, R. S., & Bhagwat, S. S. (2005). Use of an artificial neural network in modeling yeast biomass and yield of β-glucan. *Process Biochemistry*, 40(5), 1617–1626. **ANN media tuning · panel ③ network**

Zhao, M., Ma, J., Zhang, L., et al. (2024). Engineering strategies for enhanced heterologous protein production by *Saccharomyces cerevisiae*. *Microbial Cell Factories*, 23, 32. **Protein engineering**